Debiased Machine Learning

Michal Kolesár

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Warm-up: Partly linear model

Standard semiparametrics

Debiased GMM

- Consider estimating θ_0 in the partly linear model, $Y = D\theta_0 + \gamma_0(X) + \epsilon$, with $E[\epsilon \mid D, X] = 0$. Model implies the moment condition $E[D(Y - D\theta_0 - \gamma_0(X))] = 0$, where $\gamma_0 = \gamma_{Y,0} - \theta_0\pi_0$, with $\gamma_{Y,0}(X) = E[Y \mid X]$ and $\pi_0(X) = E[D \mid X]$.
- Suppose we have some preliminary estimates $\hat{\gamma}_Y$ and $\hat{\pi}$. Plugging them into the moment condition gives the sample moment condition

$$E_n[D(Y - D\theta_0 - \hat{\gamma}_Y(X) + \theta_0\hat{\pi}(X))] = 0, \tag{1}$$

and the plug-in estimator

$$\hat{\theta}_{PI} = \frac{E_n[D(Y - \hat{\gamma}_Y(X))]}{E_n[D(D - \hat{\pi}(X))]}.$$

- Notation: $\tilde{\pi} = \hat{\pi} - \pi_0$, and $\|\pi\| = E[\pi(X)^2]^{1/2}$.
- Writing $\hat{\gamma} = \hat{\gamma}_Y - \theta_0 \hat{\pi}$, the plug-in estimator satisfies

$$\hat{\theta}_{PI} - \theta_0 = \frac{E_n[D(Y - D\theta_0 - \hat{\gamma}(X))]}{E_n[D(D - \hat{\pi}(X))]} = \frac{E_n[D\epsilon] + E_n[-D\tilde{\gamma}(X)]}{E_n[D(D - \hat{\pi}(X))]}$$

- Denominator converges to $E[(D - \pi(X))^2] = E[\operatorname{var}(D \mid X)]$, provided that $\|\tilde{\pi}\| = o_p(1)$.
- First numerator term standard, mean of second term potentially non-zero, representing bias:

$$B = E[-D\tilde{\gamma}(X)].$$

We need it to be of order $o_p(1/\sqrt{n})$. Needs to be analyzed on a case-by-case basis.

The power of series

- Suppose we use a series estimator, so that $\hat{\gamma}(X) = X'\hat{\delta}$, where $\hat{\delta}$ is obtained by regressing $Y - D\theta_0$ onto X.
- This "series estimator" implies $\hat{\theta}_{PI}$ is just ordinary least squares (OLS) from regressing Y onto D and X (but we think of X as a series expansion of some baseline set of controls).
- Write $\gamma_0(X) = X'\delta + r_\gamma(X)$ and $\pi_0(X) = X'\rho + r_\pi(X)$. Then, letting H denote the projection matrix associated with X, $\tilde{\gamma}(X_i) = e_i'H\epsilon - r_\gamma(X_i) + e_i'Hr_\gamma$, so that

$$B = -r_\pi'(I - H)r_\gamma \preceq \|r_\pi\|\|r_\gamma\|.$$

- So series is doubly robust in that its bias equals the product of the two approximation biases (as we discussed in OLS lecture).

- What if we use something fancier, say the lasso, or use Akaike information criterion (AIC) to drop some elements of X? Suppose we sample split. Then

$$B = E[-\pi(X)\tilde{\gamma}(X)].$$

We run into trouble if estimation error $\tilde{\gamma}(X)$ correlated with propensity score.

  - Suppose $\gamma(X) = X\delta$, with X scalar, and $\pi(X) = X\rho$. Using $\hat{\gamma}(X) = 0$ dominates the use of $X\hat{\delta}$ in mean squared error (MSE) if $\delta^2 < \sigma_\epsilon^2/nE[X^2]$. Leads to $B \asymp 1/\sqrt{n}$ if, say, $\delta^2 = \sigma_\epsilon^2/2nE[X^2]$, but ρ is a non-zero constant.
  - Point: for picking controls X_j, we don't care about δ_j^2, but about $\rho_j\delta_j$. See Chernozhukov et al. (2022, Appendix C) and Leeb and Pötscher (2005).

Double machine learning to the rescue

- Under sample splitting, can write $B = E[\pi(X)(Y - D\theta_0 - \hat{\gamma}(X))]$, and estimate bias as $E[\pi(X)(Y - D\theta_0 - \hat{\gamma}_Y(X) + \theta_0\hat{\pi}(X))]$. Use this to debias (1), obtaining $E_n[(D - \hat{\pi}(X))(Y - D\theta_0 - \hat{\gamma}_Y(X) + \theta_0\hat{\pi}(X))]$, and the debiased estimator

$$\hat{\theta}_{DB} = \frac{E_n[(D - \hat{\pi}(X))(Y - \hat{\gamma}_Y(X))]}{E_n[(D - \hat{\pi}(X))^2]}.$$

- Population analog of the debiased moment condition is doubly robust:

$$E[(D - \pi_0(X))(Y - D\theta_0 - \gamma_0(X))] = 0.$$

- Now bias given by $B = E[\tilde{\pi}(X)\tilde{\gamma}(X)] \leq \|\tilde{\pi}\|\|\tilde{\gamma}\|$. We are OK so long as our estimators of π_0 and γ_0 converge faster than $n^{-1/4}$.

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General semiparametric setup

- A plethora of causal/structural parameters in economics can be expressed via semiparametric moment conditions $E[g(Z, \gamma_0, \theta_0)] = 0$, where γ_0 is a nuisance function (conditional mean, quantile, density etc), and θ_0 parameter of interest.
- Focus on case with
  1. $\gamma_0(X) = E[Y \mid X]$
  2. $g(Z, \gamma, \theta) = m(Z, \gamma) - \theta$, and $m(Z, \gamma) = m(Z, 0) + m_\Delta(Z, \gamma)$, with m_Δ linear in γ, and $|m_\Delta(Z, \gamma)| \leq K\|\gamma\|$

  m_Δ linear in γ allows for estimation based on doubly robust moments, rather than just locally robust/orthogonal moments (Chernozhukov et al. 2022), and simpler regularity conditions, but most theory goes through without these simplifications

$\theta_0 = E[m(Z, \gamma_0)]$ where $\gamma_0(X) = E[Y \mid X]$

- Running example: $X = (D, W)$, and $\theta_0 = E[\gamma_0(1, W)]$
    - Under unconfoundedness (i.e. $E[Y(1) \mid D = 1, W] = E[Y(1) \mid W]$), $\theta_0 = E[Y(1)]$.
    - Inference on the ATE just slightly more complicated, here $\theta_0 = E[\gamma_0(1, W) - \gamma_0(0, W)]$.
- Other examples: Average effect of shifting distribution of covariates $\theta_0 = \int \gamma_0(x) dF_1(x) - E[\gamma_0(X)]$ (Stock 1989), average derivative $\theta_0 = E[\partial\gamma_0(D, W)/\partial D]$

  …hundreds of papers applying general theory to different settings
- We will see later setup also covers partly linear model as special case, as well as instrumental variables (IV).

Plug-in approach

- Simplest approach: estimate γ_0 nonparametrically, and then set $\hat{\theta}_{PI} = E_n[m(Z, \hat{\gamma})] = E_n[\hat{\gamma}(1, W)]$.
- Then
$$\hat{\theta}_{PI} - \theta_0 = E_n[\hat{\gamma}(1, W) - \gamma_0(1, W)] + E_n[\gamma_0(1, W) - \theta_0].$$
  Second term well-behaved and asymptotically normal (corresponds to an oracle), but first typically non-zero mean, since $\hat{\gamma}$ biased.
- Need bias to be $o_p(n^{-1/2})$, can be hard to ensure (typically check on case-by-case basis), and doesn't hold if e.g. $\hat{\gamma}$ is estimated via the lasso $\implies$ $\hat{\theta}_{PI}$ asymptotically biased.

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Debiasing: Key insight!

- Notice that for any γ,

$$E[\gamma(1,W)] = E\left[\frac{D}{\pi(W)}\gamma(X)\right] \implies E[\gamma(1,W) - \gamma_0(1,W)] = E\left[\frac{D}{\pi(W)}(\gamma(X) - Y)\right].$$

  where $\pi(W) = E[D \mid W]$ is propensity score.

- Suggests estimating bias as $E_n[\frac{D}{\pi(W)}(\hat{\gamma}(X) - Y)]$
- If $\hat{\gamma}$ estimated in separate sample and $\pi(W_i)$ known, then subtracting off $E_n[\frac{D}{\pi(W)}(\hat{\gamma}(X) - Y)]$ would yield estimator of θ_0 that is unbiased conditional on $\hat{\gamma}$.

Debiased GMM

- Debiased generalized method of moments (GMM) subtracts off feasible bias estimate,

$$\hat{\theta}_{DB} = \frac{1}{n}\sum_{i=1}^{n}\hat{\gamma}_{\ell(i)}(1, W_i) + \frac{1}{n}\sum_{i=1}^{n}\hat{\alpha}_{\ell(i)}(X_i)(Y_i - \hat{\gamma}_{\ell(i)}(X_i))$$

  where $\hat{\gamma}_{\ell(i)}$ estimated on sample excluding i and $\hat{\alpha}$ is some estimator of $\alpha(X) := D/\pi(W)$.
- To avoid estimating γ_0 n times, usually split sample into $L = 5$ or $L = 10$ folds, and $\ell(i)$ excludes fold that i belongs to.
- This cross-fitting avoids own observation bias, analogous to the jackknife instrumental variables estimator (JIVE) in IV settings.

General theory

- Since $\theta(\gamma) = E[m(Z,0)] + E[m_\Delta(Z,\gamma)]$ is an affine functional, if the functional is continuous (i.e. $E[m_\Delta(Z,\gamma)] \leq C\|\gamma\|$ for all $\gamma \in L^2$ as we've assumed above), by the Riesz representation theorem, there exists $\alpha_0 \in L^2$ such that for all $\gamma \in L^2$

$$E[m_\Delta(Z,\gamma)] = E[\alpha_0(X)\gamma(X)]. \tag{2}$$

  Called Riesz representer (RR). Existence of RR equivalent to semiparametric variance bound being finite (Hirshberg and Wager 2021; Newey 1994), i.e. $\sqrt{n}$-consistent estimation possible

- In our case, $\alpha(X) = D/\pi(W)$. General recipes for RR construction available (e.g. Chernozhukov et al. 2022; Chernozhukov et al. 2018)

- Debiased GMM estimator $\hat{\theta}_{DB} = \frac{1}{n}\sum_{i=1}^{n}[m(Z_i, \hat{\gamma}_{\ell(i)}) + \hat{\alpha}_{\ell(i)}(X_i)(Y_i - \hat{\gamma}_{\ell(i)}(X_i))]$ equivalent to GMM based on moment condition

$$\theta_0 = E[\psi(Z, \gamma_0, \alpha_0)], \quad \psi(Z, \gamma, \alpha) = m(Z, \gamma) + \alpha(X)(Y - \gamma(X))$$

In our case $\psi(Z, \gamma, \alpha) = \gamma(1, W) + (Y - \gamma(X))D/\pi(W)$.

- Moment condition doubly robust

$$\begin{aligned} E[\psi(Z, \gamma, \alpha)] - \theta_0 &= E[m(Z, \gamma) - m(Z, \gamma_0) + \alpha(X)(Y - \gamma(X))] \\ &=_{(1)} E[m_\Delta(Z, \gamma - \gamma_0) + \alpha(X)(\gamma_0(X) - \gamma(X))] \\ &=_{(2)} E[(\alpha(X) - \alpha_0(X))(\gamma_0(X) - \gamma(X))] \end{aligned}$$

where (1) uses definition of γ_0 and (2) uses definition of RR, $E[m_\Delta(Z, \gamma)] = E[\alpha_0(X)\gamma(X)]$.

Doubly robust moments

- Idea of using doubly robust moments old (Robins, Rotnitzky, and Zhao 1994, 1995). Innovation of debiased GMM is to combine it with cross-fitting, which allows for weaker regularity conditions on estimator of γ that accommodate regularized estimators (lasso, random forests, neural nets etc).
- Existence of RR allows for multiple approaches to estimating θ_0: regression (plug-in), propensity score weighting, and doubly-robust moments:

$$\theta_0 = E[m(Z, \gamma_0)] = E[\alpha_0(X)Y] = E[\psi(Z, \gamma_0, \alpha_0)].$$

The following result is a variant of Theorem 1 in Chernozhukov, Newey, and Singh (2022) and Lemma 8 in Chernozhukov et al. (2022) and Lemma 6 in Newey and Robins (2018):

Proposition

Suppose that $E[m_\Delta(Z_i, \gamma)^2] \leq K\|\gamma\|^2$, $\mathrm{var}(Y_i \mid X_i) \leq K$, and that α_0 is bounded (strong overlap). Then

$$\sqrt{n}(\hat{\theta}_{DB} - \theta_0) = \mathcal{N}(0, \mathrm{var}(\psi(Z, \gamma_0, \alpha_0))) + R_n,$$

where $R_n \precsim L^{-1/2} \sum_{\ell=1}^{L}(\|\tilde{\alpha}_\ell\| + \|\tilde{\gamma}_\ell\|) + L^{-1} \sum_{\ell=1}^{L} \sqrt{n}\|\tilde{\alpha}_\ell\|\|\tilde{\gamma}_\ell\|$. In particular, if L is fixed, R_n is negligible if $\|\tilde{\alpha}_\ell\| + \|\tilde{\gamma}_\ell\| = o_p(1)$ and $\sqrt{n}\|\tilde{\alpha}_\ell\|\|\tilde{\gamma}_\ell\| = o_p(1)$.

Proof

Decompose, letting $\tilde{\alpha}_\ell = \hat{\alpha}_\ell - \alpha_0$,

$$\hat{\theta}_{DB} - E_n[\psi(Z, \gamma_0, \alpha_0)] = \frac{1}{n}\sum_{i=1}^{n}(m_\Delta(Z_i, \tilde{\gamma}_{\ell(i)}) + \hat{\alpha}_{\ell(i)}(X_i)(Y_i - \hat{\gamma}_{\ell(i)}(X_i)) - \alpha_0(X_i)(Y_i - \gamma_0(X_i))) = \sum_{\ell=1}^{L}(R_{1\ell} + R_{2\ell} + R_{3\ell}),$$

where

$$R_{1\ell} = \frac{1}{n}\sum_{i\in\ell}[m_\Delta(Z_i, \tilde{\gamma}_\ell) - \alpha_0(X_i)\tilde{\gamma}_\ell(X_i)], \qquad R_{2\ell} = \frac{1}{n}\sum_{i\in\ell}\tilde{\alpha}_\ell(X_i)(Y_i - \gamma_0(X_i))$$

$$R_{3\ell} = -\frac{1}{n}\sum_{i\in\ell}\tilde{\alpha}_\ell(X_i)\tilde{\gamma}_\ell(X_i).$$

$R_{2\ell}$ is mean zero conditional on I_ℓ^c, with variance bounded by $\max_i \operatorname{var}(Y_i \mid X_i) \cdot \|\tilde{\alpha}_\ell\|^2/nL$. $R_{1\ell}$ is likewise mean zero with variance bounded by a constant times $E[m_\Delta(Z_i, \tilde{\gamma}_\ell)^2 + \alpha_0(X_i)^2\tilde{\gamma}_\ell(X_i)^2 \mid I_\ell^c] \preceq (K + \max_i \alpha_0(X_i)^2)\|\tilde{\gamma}_\ell\|^2/nL$. Thus, by Markov's inequality the first two terms are of the order $(\|\tilde{\alpha}_\ell\| + \|\tilde{\gamma}_\ell\|)/\sqrt{nL}$. Finally, by Cauchy-Schwarz and Markov's inequality, $R_{3\ell}$ is of the order $\|\tilde{\alpha}_\ell\|\|\tilde{\gamma}_\ell\|/L$. Summing over the partitions yields the result.

Key advantage of orthogonality

- Only product of biases needs to be small!: $\sqrt{n}||\tilde{\alpha}_\ell||||\tilde{\gamma}_\ell|| = o_p(1)$. Allows for model selection / machine learning.
- In contrast, with plug-in approach, a general sufficient condition is that bias of $\hat{\gamma}$ is small, $\sqrt{n}||\tilde{\gamma}_\ell|| = o_p(1)$.
- But need to estimate α as well (which was for free in the partially linear model). It could be that α is non-smooth and hard to estimate.

Series is doubly robust

The following result is a variant on Theorem 4 in Newey and Robins (2018), and on Theorem 8 and Corollary 9 in Ichimura and Newey (2017):

Proposition

Suppose $E[m_\Delta(Z_i, \gamma)^2] \leq K\|\gamma\|^2$ and $\mathrm{var}(Y_i \mid X_i) \leq K$. For the cross-fitted plug-in estimator based on splines,

$$\sqrt{n}(\hat{\theta}_{PI} - \theta_0) = \mathcal{N}(0, \mathrm{var}(\psi(Z, \gamma_0, \alpha_0))) + R_n,$$

where the remainder term satisfies $R_n \preceq \sqrt{p \log(p)/n}(1 + \sqrt{p}\|r_\gamma\|) + \Delta_n^*$. Here $\Delta_n^* = \sqrt{p/n} + \|r_\alpha\| + \|r_\gamma\| + \sqrt{n}\|r_\alpha\|\|r_\gamma\|$.

Proof i

Let $\hat{\gamma}_\ell(x_i) = x_i'\hat{\delta}_\ell$, $\tilde{\delta}_\ell = \hat{\delta}_\ell - \delta$, and let $\epsilon_i = Y_i - \gamma_0(X_i)$. Normalize $E[XX'] = I$. Then we can decompose $\hat{\theta}_{PI} - E_n[\psi(Z, \gamma_0, \alpha_0)] = \frac{1}{n}\sum_\ell (R_{1\ell} + R_{2\ell})$, where

$$R_{1\ell} = \sum_{i\in\ell}(m_\Delta(Z_i, \tilde{\gamma}_\ell) - E[\alpha_0(X)\tilde{\gamma}_\ell(X)]), \qquad R_{2\ell} = \sum_{i\in\ell}(E[\alpha_0(X)\tilde{\gamma}_\ell(X)] - \alpha_0(X_i)\epsilon_i).$$

The term $R_{1\ell}$ is mean zero conditional on I_ℓ^c, with variance bounded by $\sum_{i\in\ell} E[m_\Delta(Z, \tilde{\gamma}_\ell)^2 \mid I_\ell^c] \le K\|\tilde{\gamma}_\ell\|^2 n/L$, so that $\sum_\ell R_{1\ell} \preceq L^{-1/2}\sum_\ell n^{-1/2}\|\tilde{\gamma}_\ell\|$ (without cross-fitting, we can bound R_{1i} using the results from Andrews 1994). Since $\|\tilde{\gamma}_\ell\| \asymp \|r_\gamma\| + \sqrt{p/n}$ by standard arguments, it follows that, with a fixed number of folds,

$$\frac{1}{\sqrt{n}}\sum_\ell R_{1\ell} \preceq \|r_\gamma\| + \sqrt{p/n}. \tag{3}$$

Proof ii

Observe that $E[\alpha_0(X)\tilde{\gamma}_\ell(X)] = E[\alpha_0(X)X\tilde{\delta}_\ell] - E[\alpha_0(X)r_\gamma(X)] = \rho_0'\tilde{\delta}_\ell - E[r_\alpha(X)r_\gamma(X)]$, using the fact that under the normalization, $\rho_0 = E[X\alpha_0(X)]$ and that r_γ is orthogonal to linear functions of X. Hence,

$$\frac{1}{n}\sum_\ell R_{2\ell} = \frac{1}{L}\sum_\ell \rho_0'\tilde{\delta}_\ell - \frac{1}{n}\sum_i \alpha_0(X_i)\epsilon_i - E[r_\alpha(X)r_\gamma(X)].$$

By Lemma 4.1 in Belloni et al. (2015),

$$\sqrt{N}\rho_0'\tilde{\delta}_\ell = N^{-1/2}\sum_{i\notin\ell}\rho_0'X_i\epsilon_i + O_p(\sqrt{p\log(p)/N}(1+\sqrt{p}\ell_p\|r_\gamma\|) + \ell_p\|r_\gamma\|),$$

with $\ell_k = 1+\log(k)$ for polynomials, while ℓ_k constant for splines, where $N = n(1-1/L)$. Therefore, with a fixed number of folds,

$$\frac{1}{L}\sum_\ell \rho_0'\tilde{\delta}_\ell = \frac{1}{n(L-1)}\sum_{\ell=1}^{L}\sum_{i\notin\ell}\rho_0'X_i\epsilon_i + O_p(\sqrt{p\log(p)}\cdot n^{-1}\cdot(1+\sqrt{p}\ell_p\|r_\gamma\|) + \ell_p\|r_\gamma\|).$$

Proof iii

Since $\sum_\ell \sum_{i \notin \ell} \rho_0' X_i \epsilon_i = (L-1) \sum_i \rho_0' X_i \epsilon_i$, it follows that for splines,

$$\begin{aligned} \frac{1}{\sqrt{n}} \sum_\ell R_{2\ell} &\asymp \frac{1}{\sqrt{n}} \sum_i r_\alpha(X_i)\epsilon_i + \sqrt{p\log(p)/n}(1+\sqrt{p}\|r_\gamma\|) + \|r_\gamma\| + \sqrt{n}\|r_\alpha\|\|r_\gamma\| \\ &\asymp \|r_\alpha\| + \sqrt{p\log(p)/n}(1+\sqrt{p}\|r_\gamma\|) + \|r_\gamma\| + \sqrt{n}\|r_\alpha\|\|r_\gamma\|. \end{aligned}$$

Combining this with (3) yields the result.

Comments

- Here Δ_n^* is the optimal order of the remainder, as discussed in Newey and Robins (2018).
- The extra term in proposition comes from not knowing the design, i.e. not knowing $E[X_i X_i']$—this is apparent from Lemma 4.1 in Belloni et al. (2015).
- With cross-fitting, series is asymptotically efficient so long as $s_\alpha + 2s_\gamma > 1$, which is a bit stronger than the minimal smoothness condition. Here we use notation $\|r_\alpha\| = p^{-s_\alpha}$, where s_α is the number of derivatives divided by the dimension of the baseline covariates
- Use series if you can!
- Propositions above also apply to the partially linear model, with $\alpha = -\pi$, and $m(Z, \gamma) = D(Y - D\theta_0 - \gamma(X))$, so that $\psi(\alpha, \gamma) = D(Y - D\theta_0 - \gamma(X)) - \pi\epsilon$, except the asymptotic variances are divided by $E[(D - \pi(X))^2]^2$.

- One can debias a plug-in estimator in many ways, see e.g., van der Laan and Rubin (2006), van der Laan and Rose (2018), Hirshberg and Wager (2021), and Athey, Imbens, and Wager (2018). I'm not clear on relative merits of different approaches.
- Cross-fitting not needed for estimating β in partially linear model $Y = D\beta + g(W) + U$ (Donald and Newey 1994)
    - This is why post double lasso (Belloni, Chernozhukov, and Hansen 2014) doesn't need to cross-fit
    - Reason for this is that there is no endogeneity, $E[Y - \gamma_0(X) \mid Z] = 0$, so no own observation bias from estimating γ_0. In general, cross-fitting not needed (at least with series!) if there is no endogeneity in this sense.
    - Otherwise, cross-fitting solves own observation bias. Generalization of jackknife IV to other settings.

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